

**THE FOURTEENTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

March 6, 1954

Morning Session

1. Let n be an odd integer greater than 1. Let A be an n by n symmetric matrix such that each row and each column of A consists of some permutation of the integers $1, \dots, n$. Show that each one of the integers $1, \dots, n$ must appear in the main diagonal of A . (page 387)

2. Consider any five points P_1, P_2, P_3, P_4, P_5 in the interior of a square S of side-length 1. Denote by d_{ij} the distance between the points P_i and P_j . Prove that at least one of the distances d_{ij} is less than $\sqrt{2}/2$. Can $\sqrt{2}/2$ be replaced by a smaller number in this statement? (page 387)

3. Prove that if the family of integral curves of the differential equation

$$\frac{dy}{dx} + p(x)y = q(x) \quad p(x) \cdot q(x) \neq 0$$

is cut by the line $x = k$, the tangents at the points of intersection are concurrent. (page 388)

4. A uniform rod of length $2k$ and weight w rests with the end A against a smooth vertical wall, while to the lower end B is fastened a string BC of length $2b$ coming from a point C in the wall directly above A . If the system is in equilibrium, determine the angle ABC . (page 389)

5. If $f(x)$ is a real-valued function defined for $0 < x < 1$, then the formula $f(x) = o(x)$ is an abbreviation for the statement that

$$\frac{f(x)}{x} \rightarrow 0 \quad \text{as } x \rightarrow 0.$$

Keeping this in mind, prove the following: if

$$\lim_{x \rightarrow 0} f(x) = 0 \quad \text{and} \quad f(x) - f\left(\frac{x}{2}\right) = o(x),$$

then $f(x) = o(x)$.

(page 390)

6. Suppose that $u_0, u_1, u_2, \dots$ is a sequence of real numbers such that

$$u_n = \sum_{k=1}^{\infty} u_{n+k}^2 \quad \text{for } n = 0, 1, 2, \dots$$

Prove that if $\sum u_n$ converges then $u_k = 0$ for all k . (page 391)

7. Prove that there are no integers x and y for which

$$x^2 + 3xy - 2y^2 = 122. \quad (\text{page 392})$$

Afternoon Session

1. Show that the equation $x^2 - y^2 = a^3$ has always integral solutions for x and y whenever a is a positive integer. (page 392)

2. Assume as known the (true) fact that the alternating harmonic series

$$(1) \quad 1 - 1/2 + 1/3 - 1/4 + 1/5 - 1/6 + 1/7 - 1/8 + \dots$$

is convergent, and denote its sum by s . Rearrange the series (1) as follows:

(2)

$$1 + 1/3 - 1/2 + 1/5 + 1/7 - 1/4 + 1/9 + 1/11 - 1/6 + \dots$$

Assume as known the (true) fact that the series (2) is also convergent, and denote its sum by S . Denote by s_k, S_k the k th partial sum of the series (1) and (2) respectively. Prove the following statements.

$$(i) \quad S_{3n} = s_{4n} + \frac{1}{2} s_{2n},$$

$$(ii) \quad S \neq s. \quad (\text{page 393})$$

3. Let a and b denote real numbers such that $a < b$. The symbol (a, b) will denote the closed interval with the end points a, b . Let there be given a collection of closed intervals $(a_1, b_1), \dots, (a_n, b_n)$ such that any two of these closed intervals have at least one point in common. Prove that there exists then a point which is contained in every one of these intervals. (page 394)

4. Given the focus f and the directrix D of a parabola P and a line L , describe (with proof) a Euclidean (i.e., ruler and compass) construction of the point or points of intersection of L and P . Be sure to identify the case for which there are no points of intersection. (page 395)

5. Let $f(x)$ be a real-valued function, defined for $-1 < x < 1$, such that $f'(0)$ exists. Let a_n, b_n be two sequences such that

$$-1 < a_n < 0 < b_n < 1, \quad \lim_{n \rightarrow \infty} a_n = 0, \quad \lim_{n \rightarrow \infty} b_n = 0.$$

Prove that

$$\lim_{n \rightarrow \infty} \frac{f(b_n) - f(a_n)}{b_n - a_n} = f'(0). \quad (\text{page 397})$$

6. Prove that every positive rational number is the sum of a finite number of distinct terms of the series

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \dots. \quad (\text{page 398})$$

7. Show that

$$\lim_{n \rightarrow \infty} \sum_{s=1}^n \left(\frac{a+s}{n} \right)^n \quad (a > 0)$$

lies between e^a and e^{a+1} . (page 399)